Stockout-Aware Demand Forecasting and Newsvendor Ordering on FreshRetailNet-50K

Akshara Kumari[1], Aman Koley[2], Shivam Singh[3], and G. Premalatha[4]

***Abstract*—Recorded sales understate demand whenever a product sells out, so forecasts and orders fitted to sales inherit the stocking policy. We study demand forecasting and newsvendor ordering on FreshRetailNet-50K, a fresh-retail panel with hourly stockout flags, where days without a flagged stockout give an exact, though selected, demand proxy. Three simulated stockout mechanisms censor these days, so models trained on differently censored histories are scored against the same uncensored outcomes on identical rows. In a 5,000-series run locked before the final week was scored, we compare nine ordering policies under three cost ratios; none wins under every ratio. When a shortage costs three times as much as a leftover unit, LightGBM demand quantiles with pooled conformal calibration are the cheapest full-sample policy in all nine train/test cells; when leftovers cost more, a raw-sales LightGBM with residual quantiles is cheapest in all nine. Training on a different mechanism raises the calibrated policy's mean cost by at most 0.0472 normalized units (95% cluster-bootstrap interval 0.0391–0.0554), the smallest worst case among the policies that read the evaluation history. For real censored days, where demand is unobserved, we report only a lower bound on misses of the 90% quantile forecast.**

***Index Terms*—censored demand, stockouts, demand forecasting, quantile regression, conformal calibration, newsvendor, fresh retail**

I. Introduction

Retailers record sales, not demand: when a product sells out, the unmet part of demand leaves no trace [1]. A model fitted to such records learns the stocking policy along with customer behavior, and in fresh retail its errors carry straight into the next order, which trades lost sales against perishable surplus [2], [3]. A lower forecast error therefore does not show that one ordering policy is better than another; the comparison needs the same outcome, the same cost ratio, and the same scored rows.

FreshRetailNet-50K [4], [5] records an hourly stock-status flag next to hourly sales. A series-day with no flagged stockout between 06:00 and 21:59 is a *donor day*, and its recorded sales over that window are an exact demand proxy, provided the flags are correct. Donor days are not a random sample, and the proxy says nothing about days that were actually censored, but they support a controlled experiment, as in the dataset report's recovery evaluation [4]. Three simulated stockout mechanisms censor the histories of donor days, features and forecasts are rebuilt from each censored history, and every forecast and order is scored against the same proxy, so a model trained under one mechanism can be evaluated under another on identical rows.

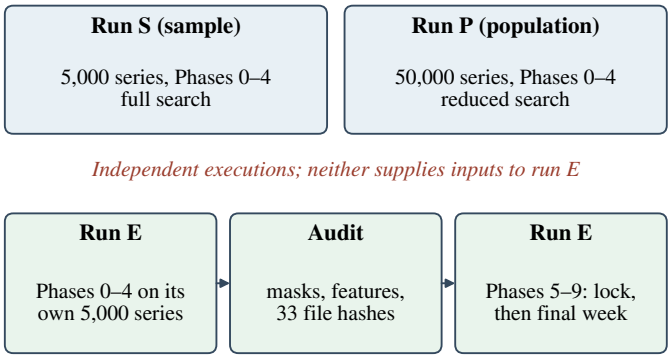


Fig. 1. The three saved executions. Only run E's own Phase 0–4 stage feeds its prerequisite audit and the locked final evaluation; runs S and P are independent development evidence.

The evidence comes from three saved executions of one notebook pipeline (Fig. 1). Two development runs prepare the data, simulate censoring, and tune the models: run S on a stratified sample of 5,000 series and run P on all 50,000 series. The end-to-end run E repeats these steps on its own 5,000-series sample and continues through neural and censored-inventory baselines, conformal calibration, ordering, a protocol lock, and a single evaluation of the final week. Only run E feeds the final evaluation.

Our contribution is an evaluation design and its evidence, not a new forecaster. Earlier work on this dataset scores demand recovery and forecast accuracy [4] or profit on censored sales [6]. We (i) score ordering cost against an uncensored proxy, comparing nine policies on identical rows; (ii) vary how the training history was censored and measure the cost of transfer between mechanisms with paired cluster-bootstrap intervals; and (iii) show that the ranking of calibrated demand quantiles and a raw-sales model reverses with the cost ratio in every train/test cell, and bound the miss rate of the 90% quantile forecast (q90) on real censored days.

Related work. Censored-demand newsvendor methods fit a parametric demand distribution [2] or learn order levels nonparametrically with the Kaplan–Meier estimator [3], [7], which our Kaplan–Meier baseline follows. Theory characterizes what censoring hides about the optimal order [8], data-driven newsvendor methods have been adapted to censored samples [9], and treating sales as demand can degrade badly as censored data

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accumulate [10]. Feature-based and neural newsvendor methods learn orders from covariates [11]–[13]; under linear costs the optimal order is a quantile, which ties ordering to quantile regression [14]. Our forecasters are LightGBM [15], used by practically all of the top 50 entries in the M5 competition [16], TimesNet [17] for recovering masked hours, and the Temporal Fusion Transformer (TFT) [18]; simple models often match transformer forecasters [19]. Calibration follows conformalized quantile regression [20], adaptive conformal inference [21], and monotone rearrangement [22], and conformal calibration has been proposed for the feature-based newsvendor [23]; its guarantees rest on exchangeability or bounds on departures from it [24], which real censored days do not meet. On this dataset, the report [4] scores recovery on synthetically masked non-stockout periods with an official baseline [25] that we reproduced only in part, and Azar [6] selects newsvendor policies but scores profit on censored sales, whose bias is not identified. Elsewhere, stockout-aware evaluation excludes out-of-stock days [26] or feeds a cost-aware ordering rule [27]. None of these scores orders against an uncensored outcome while varying the censoring mechanism.

II. Data and Experimental Protocol

A. Data, outcome, and split

The training file has 4,500,000 series-day rows for 50,000 (city, store, product) series, and the evaluation file has 350,000. The local files hold 865 product identifiers, as on the current dataset card [5]; the dataset report states 863 stock-keeping units [4]. We use release 1.0 (CC BY 4.0); its upstream revision and our download date were not recorded, so the SHA-256 hashes in the availability statement identify the files. The outcome is the recorded sales from 06:00 to 21:59, the window a next-day order has to cover. A series-day is a donor if its stock-status flags $s_{j,t,h}$ sum to zero over that window, and only then are its window sales the proxy:

$$I^{\text{donor}}_{j,t} = \mathbf{1}\Big\{\sum_{h=6}^{21} s_{j,t,h} = 0\Big\}, \quad D^{\text{proxy}}_{j,t} = \sum_{h=6}^{21} y^{\text{sale}}_{j,t,h}. \tag{1}$$

The selection is measurable: in run S, fit-period donor days had a pre-day two-week mean of recorded sales 0.91 times that of the other days and fell less often on promotion days (0.35 against 0.39). Real final-week rows with a flagged stockout have no demand label and are never scored for error or cost.

The pipeline has ten phases: data contract and sampling (Phase 0), audit and past-only features (1), simulated censoring (2), conventional baselines (3), LightGBM demand quantiles (4), TimesNet and TFT (5), Kaplan–Meier (6), conformal calibration (7), ordering and the protocol lock (8), and final evaluation (9). All runs share one chronological split: days 1–60 (2024-03-28 to 2024-05-26) fit the models, days 61–75 (to 2024-06-10) select hyperparameters, days 76–90 (to 2024-06-25) are used for calibration only, and days 91–97 (2024-06-26 to 2024-07-02) form the final week. Sampling tiers (first-level category, sales tier, and stockout-rate tier) and the initial simulator rules use fit-period data only. Features use only earlier information: target-day flags, sales, weather, and promotions are excluded, lags are shifted by at least one day, and randomizing every target-day field leaves the features unchanged. Until Phase 4 the evaluation file was read only for its schema, identifiers, and dates; run E wrote its protocol lock in Phase 8, before any final-week outcome was scored, and the final week has been used once.

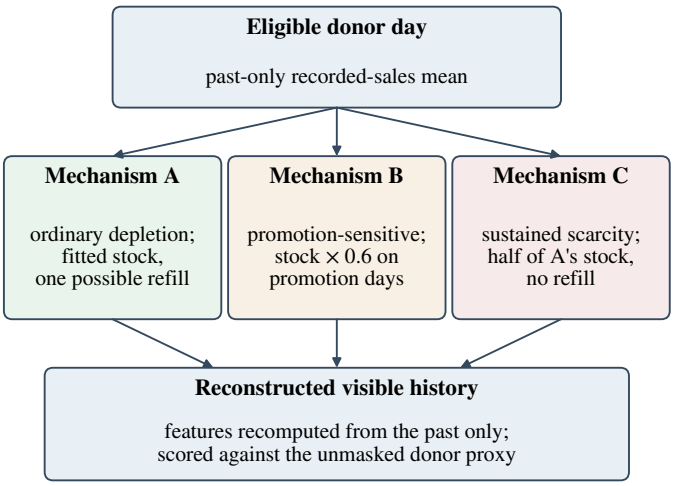


Fig. 2. The three censoring mechanisms act on eligible donor days. Each produces its own visible history, from which past-only features and forecasts are rebuilt; real censored days keep their recorded values.

B. Simulated censoring mechanisms

Each mechanism replaces the visible history of eligible donor days, those with enough past data, by a simulated censored history; real censored days keep their recorded values. The opening stock of series j on day t under mechanism m is

$$Q^{(m)}_{j,t} = \max\Big\{\mu^{\text{past}}_{j,t}\, k^{(m)}_{j,t} \exp(\sigma_m z_{j,t}),\ q_{\text{floor}}\Big\}, \tag{2}$$

where $\mu^{\text{past}}_{j,t}$ is the mean recorded window sales over the preceding two weeks, $k^{(m)}_{j,t}$ a stock multiplier fitted on fit-period stockout patterns, $z_{j,t}$ a seeded standard normal draw shared by all mechanisms, and q_{floor} a small floor; visible hourly sales are the lesser of proxy sales and remaining stock. Mechanism A (ordinary depletion) may add one refill of half the opening stock one to three hours after the first simulated stockout, with a probability estimated from real within-day recoveries. Mechanism B (promotion-sensitive) multiplies A's stock factor by 0.6 on recorded promotion days, which drive the simulator but are not a forecasting feature. Mechanism C (sustained scarcity) halves A's multiplier, uses a fixed noise scale, and never refills (Fig. 2). Within a transfer cell, defined by a training mechanism and an evaluation history, every forecast is scored against the same proxy, so cost differences are paired.

C. Models, calibration, ordering, and metrics

The conventional baselines are a seasonal-naive rule and LightGBM [15] point and quantile models trained on visible sales. The demand-quantile family trains LightGBM quantile

models on donor-proxy targets at $\tau \in \{0.25, 0.50, 0.75, 0.90\}$, one set per mechanism. On a declared deep subset, TimesNet recovers artificially hidden hours and TFT forecasts the next day from the visible (direct) or the TimesNet-recovered history. The Kaplan–Meier policy estimates an order percentile from censored sales within pools that fall back from category by sales tier to category, management group, and a global pool. Pooled conformal calibration shifts each quantile by a residual quantile of calibration-window donors in the same pool hierarchy, with a minimum pool size of 200, and then sorts each row's corrected quantiles; an adaptive variant [21] serves as a sensitivity analysis. Point forecasters become quantile forecasts by adding the pooled quantile of their calibration residuals.

With a cost c_u per unit short and c_o per unit left over, the newsvendor order is the $\beta = c_u/(c_u + c_o)$ quantile, and a row costs

$$C(Q, D^{\text{proxy}}) = c_o \max(Q - D^{\text{proxy}}, 0) + c_u \max(D^{\text{proxy}} - Q, 0) \tag{3}$$

against the unmasked proxy under a one-day shelf life. The locked settings $(c_u, c_o, \beta) = (1, 3, 0.25)$, $(1, 1, 0.50)$, and $(3, 1, 0.75)$ are called excess stock more costly, equal costs, and shortage more costly; costs are in normalized sales units. On donor rows we also report the weighted absolute percentage error (WAPE) of the q50 forecast, the empirical coverage of each quantile, and the pinball loss [14]; forecast error, coverage, and cost answer different questions, and we do not rank methods across them. For a real final-week row with recorded window sales S_{rec} and q90 forecast U_{90}, demand is at least S_{rec}, so every row with $S_{\text{rec}} > U_{90}$ is a certain miss and the definite-miss rate

$$L_{\text{miss}} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{S_{\text{rec},i} > U_{90,i}\} \tag{4}$$

is a lower bound on the real q90 miss rate, neither a coverage estimate nor an estimate of censored demand.

III. Development Runs and Model Training

Run S searched hyperparameters fully on 5,000 series; run P used a reduced search on all 50,000 series. Their overall donor shares are almost equal (0.5580 and 0.5573), and their simulator summaries agree to within 0.001: the masked share of eligible donor days is 0.460, 0.566, and 0.832 for A, B, and C in run S. Run P has a lower q50 tuning WAPE (0.3124–0.3161 against 0.3334–0.3361 over the three mechanisms) and a longer runtime, but the runs differ in search mode as well as size, so neither difference can be attributed to scale.

Run E did not reuse run S. It executed its own Phase 0–4 stage, with the full search, in the same kernel as its later phases. Its fingerprints of the selected series, the split, and the three simulator masks equal those of run S, and so do its donor counts and q50 tuning WAPEs at full stored precision (0.333440, 0.336140, and 0.335759), consistent with a deterministic pipeline; timestamps, pandas versions, and file hashes differ, so we treat the runs as separate evidence. One Phase 0 check of run E stayed open because it compares fingerprints with an earlier execution in the same output folder, and the first execution has none; the comparison with run S is the one it would have made. A prerequisite audit then passed 10 checks: it re-derived the simulator masks and feature matrices from the raw panel with exact agreement, verified the hashes of 33 predecessor files, and confirmed that the calibration and final-week days were unused. Run E's later phases ran in the reduced mode with 200 bootstrap replicates, and all results from here on come from its 5,000 series. The neural models ran on CPU with two logical CPUs and four training epochs per model on a declared subset of 600 series, so they are budget-limited references that we compare only with each other. TimesNet scored 290,397 artificially hidden donor hours with WAPE 0.917 to 0.934 across the mechanisms; TFT's next-day WAPE on 5,407 matched tuning donor days falls from 0.451–0.453 with the visible history to 0.413–0.415 with the TimesNet-recovered history. The two tasks differ in target and unit, and neither is comparable with the Phase 4 WAPE. Our reproduction of the official baseline [25] is partial: the as-documented TimesNet run completed only after a recorded patch for a blocked network download, and the upstream TFT prediction step was not run.

The Kaplan–Meier policy passed all 8 of its unit tests and built 207 pools; when a pool cannot support the requested percentile, the order is left unestimated. Calibration used 513,276 calibration-window residuals, 0.061 of the calibration rows were corrected from a pool coarser than category by sales tier, and sorting removed every remaining quantile crossing, which guarantees ordered quantiles, not calibrated ones. All 10 hand-computed checks of the order and cost arithmetic passed, and the Phase 8 lock then fixed the policy names, cost settings, model and calibration files, planned comparisons, and bootstrap settings before Phase 9 read any final-week outcome.

IV. Locked Final-Week Evaluation

The final evaluation scores donor rows only. Each of the nine transfer cells, one per pair of training mechanism and evaluation history, holds the same 20,641 final donor series-dates, on which the six full-sample policies other than Kaplan–Meier are scored. Kaplan–Meier scores only rows with an estimable order, between 6,012 and 20,641 per cell, and the two neural policies score the 2,510 final donor rows of the deep subset; no predictions are imputed for the remaining 4,400 series.

A. Policy comparison

Table I gives the mean costs of the A-trained models on the A history. The best full-sample policy depends on the cost ratio: the raw-sales LightGBM with residual quantiles is cheapest when excess stock is more costly and, narrowly, at equal costs (0.3868 against 0.3891 for pooled conformal), whereas the pooled-conformal policy is cheapest when shortage is more costly. The B history gives the same ranking; on the C history the pooled-conformal policy is already the cheaper of the two at equal costs (0.4094 against 0.4411). Kaplan–Meier is the most expensive full-sample policy in every setting, and within the deep subset TFT on the recovered history costs less than direct TFT in every matched history and setting.

TABLE I

MEAN NEWSVENDOR COST OF EVERY LOCKED POLICY FOR A-TRAINED MODELS ON THE A HISTORY (LOWEST COST PER PANEL AND SETTING IN BOLD)

Policy	Population	n	Mean cost $\beta = 0.25$	Mean cost $\beta = 0.50$	Mean cost $\beta = 0.75$
(a) Full-sample exact-donor candidates					
Seasonal naive + residual quantile	Full sample	20,641	0.6759	0.4430	0.7853
Raw-sales LightGBM + residual quantile	Full sample	20,641	**0.5448**	**0.3868**	0.7366
Raw-sales quantile LightGBM	Full sample	20,641	0.5749	0.3999	0.7503
Demand-quantile LightGBM (uncalibrated)	Full sample	20,641	0.5816	0.3946	0.7152
Demand-quantile LightGBM + pooled conformal	Full sample	20,641	0.5766	0.3891	**0.6831**
Kaplan–Meier order percentile	Full sample	20,641	0.8404	0.6520	1.3878
Demand-quantile + adaptive calibration (sensitivity)	Full sample	20,641	0.5815	0.3932	0.6963
(b) Declared deep subset of 600 series (separate population)					
TFT (direct) + residual quantile	Deep subset	2,510	0.6892	0.4949	0.9483
TimesNet→TFT + residual quantile	Deep subset	2,510	**0.6586**	**0.4687**	**0.8759**

TABLE II

PRE-SPECIFIED PAIRED CONTRASTS FOR THE POOLED-CONFORMAL POLICY OVER THE THREE MATCHED HISTORIES (NEGATIVE VALUES FAVOR POOLED CONFORMAL)

Pooled conformal minus	Cost setting	Population	n	Difference	95% percentile interval
Uncalibrated demand quantile	Excess stock more costly	Matched full sample	61,923	−0.0052	[−0.0068, −0.0034]
Uncalibrated demand quantile	Equal costs	Matched full sample	61,923	−0.0077	[−0.0096, −0.0062]
Uncalibrated demand quantile	Shortage more costly	Matched full sample	61,923	−0.0420	[−0.0483, −0.0365]
Raw-sales LightGBM + residual	Excess stock more costly	Matched full sample	61,923	+0.0208	[0.0016, 0.0432]
Raw-sales LightGBM + residual	Equal costs	Matched full sample	61,923	−0.0091	[−0.0204, 0.0052]
Raw-sales LightGBM + residual	Shortage more costly	Matched full sample	61,923	−0.0790	[−0.0946, −0.0615]
Seasonal naive + residual	Excess stock more costly	Matched full sample	61,923	−0.1213	[−0.1465, −0.0876]
Seasonal naive + residual	Equal costs	Matched full sample	61,923	−0.0694	[−0.0872, −0.0439]
Seasonal naive + residual	Shortage more costly	Matched full sample	61,923	−0.1376	[−0.1723, −0.0902]
Kaplan–Meier percentile	Excess stock more costly	KM-supported rows	61,523	−0.4043	[−0.4276, −0.3824]
Kaplan–Meier percentile	Equal costs	KM-supported rows	58,183	−0.3223	[−0.3429, −0.3046]
Kaplan–Meier percentile	Shortage more costly	KM-supported rows	47,294	−0.7475	[−0.8175, −0.6812]

TABLE III

TRANSFER GAPS OF THE POOLED-CONFORMAL POLICY WHEN SHORTAGE IS MORE COSTLY

Train	Test history	Population	n	Matched	Unseen	Gap	95% percentile interval
A	B	Full sample	20,641	0.6831	0.6932	+0.0101	[0.0076, 0.0128]
A	C	Full sample	20,641	0.6831	0.7303	+0.0472	[0.0391, 0.0554]
B	A	Full sample	20,641	0.6940	0.6908	−0.0032	[−0.0059, −0.0006]
B	C	Full sample	20,641	0.6940	0.7306	+0.0365	[0.0275, 0.0447]
C	A	Full sample	20,641	0.7248	0.7120	−0.0128	[−0.0219, −0.0050]
C	B	Full sample	20,641	0.7248	0.7086	−0.0162	[−0.0242, −0.0083]

Table II gives the paired contrasts that the lock pre-specified, pooled over the three matched histories. Calibration lowers cost relative to the uncalibrated demand quantiles in all three settings, and the pooled-conformal policy is cheaper than the seasonal-naive and Kaplan–Meier policies throughout. Against the raw-sales LightGBM with residual quantiles the sign depends on the cost ratio: pooled conformal costs more when excess stock is more costly, the equal-cost interval contains zero, and it costs less when shortage is more costly. Moving from $\beta = 0.25$ to $\beta = 0.75$ raises the pooled-conformal fill rate on the A history from 0.614 to 0.891 and the leftover share from 0.056 to 0.226, and the C history is the most expensive in every setting. Calibration moves the q90 coverage of the 20,641 matched rows from 0.798–0.828 to 0.879–0.887 and lowers the q90 pinball loss from 0.119–0.131 to 0.109–0.114; coverage stays below the nominal 0.90 and says nothing about censored days.

B. Mechanism transfer and robustness

Table III fixes the pooled-conformal policy at $\beta = 0.75$ and varies the history used for training and calibration. The transfer gap $\Delta_{m\to h} = \overline{C}_{m\to h}(\mathcal{I}) - \overline{C}_{m\to m}(\mathcal{I})$ compares the mean cost on an unseen evaluation history h with the matched cost on the same rows $\mathcal{I}$. The largest gap arises for models trained on A and evaluated on C: 0.7303 against 0.6831, a gap of +0.0472. The C-trained models cost less on the A and B histories than on their own, the hardest one, so a negative gap signals an easier history rather than better transfer. All intervals are 95% percentile intervals from 200 cluster-bootstrap replicates [28]

over the 866 city–store clusters, with the same draws for both sides of a contrast; dates are not resampled separately and no multiplicity adjustment is applied, so we read the intervals as descriptive.

The reversal holds in all nine train/test cells. When excess stock is more costly, the raw-sales LightGBM with residual quantiles is the cheapest full-sample policy in every cell, with pooled conformal 0.0007 to 0.0323 more expensive; when shortage is more costly, pooled conformal is the cheapest in every cell, 0.0441 to 0.1298 below the raw-sales policy; at equal costs four cells go to each and one to the uncalibrated demand quantiles. Over the three settings, the largest transfer gap of pooled conformal, +0.0472, is the smallest worst case among the six policies whose forecasts read the evaluation history; the raw-sales quantile LightGBM reaches +0.2258, and Kaplan–Meier orders do not read that history. Over the seven final dates, with 2,507 to 3,385 donor rows each, the C history is the most expensive on every date, and the daily q90 coverage of pooled conformal ranges from 0.839 to 0.907. Under-coverage concentrates in series with at most ten complete fit-period days (0.752 to 0.769 over 532 rows) and on promotion days (0.854 to 0.865). These breakdowns are descriptive, not pre-specified.

C. Real recorded histories

Applied to the same 35,000 real final-week rows, including rows with flagged stockouts, the q90 forecasts of the A-, B-, and C-trained pooled-conformal models give definite-miss lower bounds (4) of 0.0895 $[0.0852, 0.0942]$, 0.0854 $[0.0808, 0.0896]$, and 0.0724 $[0.0684, 0.0766]$. A smaller bound can come from a higher forecast, so it does not by itself indicate better calibration. The bound is highest for rows with 1–4 recorded stockout hours (0.1465 to 0.1797) and lowest for 9–16 hours (0.0298 to 0.0334): when a product sells out early, recorded sales seldom exceed the forecast even when demand does. Only the complete-day stratum has an exact proxy.

V. Discussion and Conclusion

The reversal with the cost ratio follows the direction of the censoring bias. Sales understate demand, which costs little when the order targets a low quantile and much when it targets a high one, so a model trained on recorded sales does well when excess stock is the expensive error and falls behind when shortage is. This qualifies the warning that treating sales as demand degrades as censored data accumulate [10]: here the penalty appears when shortage is costly. The ablations recorded in run E agree. Averaged over the matched histories at equal costs, the pooled-conformal demand quantiles cost 0.3984 against 0.4075 for the raw-sales point model; the adaptive variant cost 0.4039, so its drift correction did not pay off in this week; a hypothetical two-day carry-over of unsold stock lowers the cost to 0.2579, which shows how much the cost level depends on the one-day shelf life; and target-day weather and promotion values, unknown when the order is placed, change the q50 WAPE only from 33.48% to 33.40%.

Limitations. Donor-day sales equal demand only on complete days, which depend on demand and stocking, so the proxy does not identify demand on censored days, where we give only a lower bound. Mechanisms A, B, and C are specified simulations, not counterfactual inventory paths, so transfer results measure sensitivity to these histories, not transport to another retailer. Kaplan–Meier scores only estimable rows, and the neural policies are budget-limited references on 600 series. Rankings depend on the locked cost ratios and the one-day shelf life, and one evaluated week cannot show seasonal effects; any later protocol evaluated on it is exploratory, while the 45,000 series outside run E's sample have never been scored. The intervals use only 200 replicates. No run hashed the notebook source it executed, and the upstream dataset revision and download date are unknown, so the local file hashes are the operative identifiers. All runs used Linux, Python 3.11.15, two logical CPUs, and 8.42 GB of RAM, run E without a GPU.

Conclusion. A matched donor-proxy design makes stockout-aware forecasting and ordering comparisons on FreshRetailNet-50K checkable: models trained on differently censored histories are scored against the same outcomes, under a protocol fixed before the final week was read. Calibrated demand quantiles gave the lowest cost in every train/test cell when shortage was more costly, and a raw-sales point model with residual quantiles did so when excess stock was more costly. Evaluating the calibrated policy on an unseen mechanism raised its cost by at most 0.0472 normalized units. These findings hold for donor days, the simulated mechanisms, and the locked cost settings; testing them on censored demand requires a new protocol, and testing them on the 45,000 unscored series requires only a new lock.

Code and Data Availability

The notebooks, the saved run outputs, and scripts that regenerate every table and figure without retraining accompany this work; run E's locked protocol is in `manifests/final_test_lock.json`. The dataset is not redistributed; it is available from Hugging Face [5] under CC BY 4.0. The SHA-256 hashes of the local files and the digest of run E's locked protocol are:

```
train.parquet  6706832db892bbae4969c19d87e07975
               d2543d2ba7d7d4756360654785de5a3d
eval.parquet   1b118840664280c6b88bffc84c80ee1f
               54c05d911e354b7599e5da10995e960e
lock digest    4ab12c4bddc06b4bb32ba8825353071b
               964ef5bfdd423bd4c876ac382180c1b6
```

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